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A progressive iterative deformation snake model for segmentation of images with multi-level complexity[☆]

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ABSTRACT

We propose in this work a new and efficient active contour technique for segmenting diverse types of images. Our approach presents a new snake deformation algorithm named Progressive Iterative Deformation (PID) method, which iteratively fits control points to construct a series of fitting curves until the object boundary is detected, without the need to solve energy minimization problems. In addition, to accommodate various image types, we propose a new weighted external energy expression that controls segmentation quality based on texture and feature extraction. This method operates over three image complexity classes, adapting to segmentation challenges and image quality. We evaluate the robustness and efficiency of the proposed PID-Snake model across three different dataset classes. Our results demonstrate that the proposed technique improves deformation complexity cost and segmentation performance. In particular, our model is effective and powerful against the difficulties presented by images such as those in the medical field.

1. Introduction

Image segmentation, or object contours extraction, is one of the most important tasks in computer vision. It consists of partitioning the image into homogeneous zones according to a given criterion. This technique finds utility in various scientific domains, such as medical image analysis, environmental science, industrial quality control, autonomous vehicles, art and design, and beyond, the ability to detect contours and segment objects holds significant importance. For instance, in medical imaging, segmentation aids in diagnosis, treatment planning, and image-guided interventions [1,2]. Similarly, in environmental science, it facilitates the monitoring of land use, forest cover, and other ecological parameters using aerial or satellite imagery [3].

Active Contours (AC) or snakes introduced by Kass et al. [4] are an example of a general technique for fitting an elastic curve or spline defined in an image domain. They possess a dynamic behavior that evolves from an initial position in the image to the object edges through an iterative process of energy minimization. The idea of the contour-based approach is to bring the points of the active contour closer to high-gradient values, while maintaining properties such as curvature and elasticity of the curve. This can result in a term of attachment to data consisting of an energy functional to be minimized that contains only edge-based terms. Based on these models, several segmentation methods were developed: parametric [5–7], geometric [8,9], and geodesic active contours [10].

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To avoid the costly computation of the original AC model, Menet et al. [5] introduced parametric B-spline snake models called B-snake, where the snake's contour structure was represented by B-spline functions to approximate deformable curves using a few control points and spline basis functions. The simplicity of modeling and the regularity properties of these curves are the advantages of such an approximation. Intensive research has been carried out on parametric B-snake and many efforts have been made to increase their robustness [11–16]. Badoual et al. [11] introduced a novel family of snakes that exhibit multiresolution properties based on an admissible subdivision mask to improve the robustness of the model in case of noise. Gagan et al. [12] proposed an automated method for optic disc segmentation in retinal fundus images, employing B-snakes. This segmentation technique involves the manipulation of the active contour through scaling, translation, and rotation. Marcos et al. [13] introduced the Deep Structured Active Contours (DSAC) model, which leverages a CNN to learn AC parameters for building instance segmentation. Because the AC model is an energy function-based approach rooted in geometric and physical principles, it relies heavily on image content for segmentation. As a result, it is highly sensitive to abrupt changes in image intensity, making convergence to object edges difficult in low quality images. In addition, the object contour may appear not only by pixel intensity changes, but also by texture changes, resulting in a diminishing energy functional that disrupts the snake process. Such limitations are prevalent in various practical scenarios, including medical imaging contexts.

While processing the deformation of the snake through solving the linear minimization system, employing iterative methods such as steepest descent, ω -Jacobi, Gauss–Seidel, and Successive Over Relaxation (SOR) techniques prove advantageous. However, despite their effectiveness in addressing certain problems, these methods exhibit limitations concerning convergence speed, parameter sensitivity, and stability. Notably, the convergence behavior of these iterative techniques is highly dependent on parameter selection, and choosing an inappropriate value can result in slow convergence or even divergence of the snake. This requires additional computational effort to determine suitable parameters, thereby complicating implementation.

The Progressive and Iterative Approximation (PIA) method, as introduced by Qi et al. [17] to uncover the progressive iterative property in the context of uniform cubic B-spline curves, is an efficient and intuitive approach to data fitting by generating a sequence of approximation curves or surfaces. Recently, the progressive iterative approximation method has gained popularity [18,19] due to its ability to avoid the complexities associated with solving large systems of linear equations in reverse engineering. It also exhibits commendable adaptability and stability. In particular, the PIA method has significant geometric implications and is much easier to implement.

In this paper, inspired by the PIA method, we introduce a new intuitive and effective deformation technique called the progressive iterative deformation method. This method iteratively adjusts control points to generate a sequence of fitting curves until the object boundary is detected, eliminating the need to solve energy minimization problems. We also introduce an interval to select the parameter to ensure the convergence of the snake. This results in a quicker and simpler deformation process without performing additional computations. On the other hand, to improve the energy performance against a wide variety of image types for accurate AC-based segmentation, we propose in this work a new energy functional expression to handle three different image quality levels by highlighting their characteristics. In fact, three main classes of image quality have been explained, for each class a set of features has been employed in the energy expression. These features highlight the textures present in the image and the object features while using texture filters or the segmentation model based on a convolutional neural network, namely U-Net. In addition, supervised parameters are used to enhance the robustness of the energy.

The remainder of this paper is organized as follows: In Section 2, we provide a detailed explanation of the proposed method, including the contributions of the external energy and the PID process of deformation. Section 3 presents the experimental setup, results, and discussion. Finally, Section 4 concludes the paper.

2. Proposed framework

The B-snake model is a B-Spline curve which consists in smoothly joining curve segments. Each curve segment represents a polynomial of a given degree k . This curve is defined as a linear combination of spline basis functions $B_{i,k}$ and a sequence of control points $P_i = (X_i, Y_i)$, where $i = 0, \dots, M$. Consequently, the parametric cubic B-snake curve can be expressed as follows [5]:

$$r(s) = (x(s), y(s)) = \left(\sum_{i=0}^M X_i B_{i,3}(s), \sum_{i=0}^M Y_i B_{i,3}(s) \right), \quad (1)$$

where the B-splines $\{B_{i,3}, i = 0, \dots, M\}$ are associated with the partition $0 = s_0 = \dots = s_3 < s_4 < \dots < s_{M+1} = \dots = s_{M+4}$ of the bounded interval $J = [0, 1]$, and s is a parameter ranging within the interval J .

We denote by $\mathbf{B}$ the collocation matrix of the B-spline functions at the knots $\{t_j\}_{j=0}^M$ in the interval J defined by:

$$\mathbf{B} = (B_{i,3}(t_j))_{\substack{i=0,\dots,M \\ j=0,\dots,M}} \quad (2)$$

To describe the snake by a closed curve, we assume that the set of control points is M-periodic, i.e. $P_{i+M} = P_i, \forall i \in \{0, \dots, M\}$. Snakes derive their name from their ability to bend and adjust like real snakes. They are widely used technique for creating elastic curves within an image region. Their dynamic behavior begins from an initial position near the object to be segmented and progressively moves to accurately detect its boundaries through an iterative process of energy functional minimization. This process continues until the equilibrium position is reached, where optimal energy is obtained. The total energy function of the B-Snake $E_{B\text{-snake}}$ can be defined by

$$E_{B\text{-snake}} = E_{int} + E_{ext}. \quad (3)$$

Table 1

Description of three levels of image complexity and corresponding feature sets.

Complexity level	Description	The features set Γ
Low (LC)	The image is of excellent quality, with the object's form emerges distinctly, defined by abrupt shadings in its intensity. However, the textures exhibit a low contrast.	$\Gamma = \emptyset$.
Medium (MC)	Shape recognition may depend more on changes in texture than changes in intensity. The images may have a moderate amount of noise, which refers to random fluctuations in pixel values.	$\Gamma = \{I * \gamma_p\}_{p=1}^K$, where $\{\gamma_p\}_{p=1}^K$ is a collection of texture filters.
Hard (HC)	The image exhibits poor quality, marked by the presence of numerous noises, artifacts, and distortions. Distinguishing the object from the background proves challenging, compounded by blurred edges. Additionally, significant variability exists in the shapes of the objects depicted, further complicating the segmentation process.	$\Gamma = \{I_p\}_{p=1}^K$, where $\{I_p\}_{p=1}^K$ denote the collection of feature maps taken from the U-Net model that has already been trained [7].

The internal energy E_{int} controls the appearance of the curve such that

$$E_{int} = \frac{1}{2} \int_0^1 \beta_1 \|r'(s)\|^2 + \beta_2 \|r''(s)\|^2 ds, \quad (4)$$

where β_1 is the elasticity factor and β_2 is the rigidity factor of the contour, making it possible to obtain more or less smooth curves, and the terms r' and r'' are respectively the first and the second derivatives of r with respect to s .

The image energy E_{img} pulls the curve towards the edges of the object, and can be expressed as follows:

$$E_{img} = -\gamma \int_0^1 \|\nabla I(r(s))\|^2 ds, \quad (5)$$

where $\nabla(I(r(s)))$ represents the gradient of the image I in $r(s)$, and γ is a parameter depending on the image.

The external energy E_{ext} is commonly defined to be equivalent to the image energy, i.e. $E_{ext} = E_{img}$.

2.1. Energy functional

To develop a highly effective and robust energy functional applicable to a variety of image types and domains, we propose in this section a new external energy expression that aims to highlight the object from the background by analyzing different textures and features present in the image. This energy term is expressed by adding a new feature energy, denoted as E_{feat} , to the image energy E_{img} . The feature energy may differs regarding the quality of the image and the appearance of the object to be segmented.

Then, we define the new external energy of the snake as follows:

$$E_{ext}(r) = E_{img}(r) + E_{feat}(r) \quad (6)$$

$$= -\gamma \int_0^1 \|\nabla I(r(s))\|^2 ds - \sum_{p=1}^K \alpha_p \int_0^1 \|\nabla \Gamma_p(r(s))\|^2 ds, \quad (7)$$

where γ and $\{\alpha_p\}_{p=1}^K$ denotes the weights used to separate the target object from its background through the image and its features $\{\Gamma_p\}_{p=1}^K$ respectively. They can be obtained by maximizing the Fisher criterion [20], which is computed once immediately after the PID-snake initialization. The weights γ and $\{\alpha_p\}_{p=1}^K$ remains unchanged during the deformation process of the snake. $\Gamma = \{\Gamma_p\}_{p=1}^K$ is a collection of K features extracted from the image. In practical use, there are generally three levels of image quality. The following table explains these levels and provides the corresponding feature set Γ :

As explained from Table 1, when the segmentation complexity is low, object edges correspond to areas in the image where intensity changes abruptly, known as regions of optimal gradient. Since AC operates on image gradients, it can generate a robust external energy. Moreover, this process may eliminate features energy E_{feat} .

In most practical general cases, the contour may appear not only by pixel intensity changes, but also by texture changes. In the case, the external energy may lead to the divergence of the snake, so an inaccurate segmentation results can be provided. It is curious to highlight the different textures present in the image by processing it through filters to build an energy that helps the snake to converge towards the boundary of the object. So, if the image is of medium complexity, the simple texture filters $\{\gamma_p\}_{p=1}^K$ can be used as Gabor filters to build an energy E_{feat} that supports the deformation of the snake.

However, to overcome the snake's divergence when dealing with more difficult images, it is necessary to dig deeper into the processing of these images. This requires the use of advanced techniques capable to extract more complex features and patterns from the image. Convolutional Neural Network (CNN) models, such as U-Net [1], can efficiently process the complexity of image data and extract meaningful information and features. Then, the use of the pre-trained feature maps $\{\Gamma_p\}_{p=1}^K$ extracted from the decoder part before the final 1×1 convolutional layer of the U-Net model can improve the segmentation performance.

The complete technique for extracting the energy functional is detailed in Algorithm 1.

Algorithm 1: Procedure for extracting external energy functional of the snake

input : Input color or gray image I
output: External energy E_{ext} of the snake

```

1 //Features set  $\Gamma = \{\Gamma_p\}_{p=1}^K$  extraction
2    $\Gamma \leftarrow \emptyset$  // Low Complexity (LC) case of  $I$ 
3    $\Gamma \leftarrow \{I * \gamma_p\}_{p=1}^K$  // Medium Complexity (MC) case of  $I$ 
4   U-Net model training and features map extraction // Hard Complexity (HC) case of  $I$ 
    $\Gamma \leftarrow \{\mathcal{I}_p\}_{p=1}^K$ 
5 // Computation of the weights  $\gamma$  and  $\{\alpha_p\}_{p=1}^K$ 
6   Initialization of ROIs : A (object), and B (background)
7   Maximizing the Fisher criterion
8 // computation of the external energy
9    $E_{img} \leftarrow -\gamma \|\nabla I\|^2$ 
10   $E_{feat} \leftarrow -\sum_{p=1}^K \alpha_p \|\nabla \Gamma_p\|^2$ 
11   $E_{ext} \leftarrow E_{img} + E_{feat}$ 
12 return ( $E_{ext}$ )

```

2.2. Progressive iterative deformation process

The success of the snake approach relies heavily to its initialization before the deformation process. The choice of the initial snake and the total energy term is very critical because it determines the quality of the segmentation. In general, it is determined by the user based on an existing understanding of the data. For more convenience, it is often drawn manually. Here, our parametric snake curve is initialized by placing $M + 1$ control points close to the object boundaries, followed by the construction of the spline curve r using Eq. (1).

Let $P^0 = \{P_i^0\}_{i=0}^M$ denote the initial control points, and $r^0(s) = (x^0(s), y^0(s))$ represent the initial snake curve, where $s \in [0, 1]$, given as:

$$r^0(s) = \sum_{i=0}^M P_i^0 B_{i,3}(s). \quad (8)$$

By computing the partial derivative of the total energy $E_{B-snake}$ with respect to the control points, we define the initial adjusting vector $\Delta^0 = \{\Delta_i^0\}_{i=0}^M$, such that:

$$\Delta_i^0 = \mu \left(\int_0^1 F(r^0(s)) B_{i,3}(s) ds + \int_0^1 \sum_{j=0}^M P_j^0 \delta_{i,j}(s) ds \right), \quad \forall i = 0, \dots, M \quad (9)$$

where

$$\delta_{i,j}(s) = \beta_1 B'_{j,3}(s) B'_{i,3}(s) + \beta_2 B''_{j,3}(s) B''_{i,3}(s),$$

μ is a parameter, and $F(\cdot) = (F_x(\cdot), F_y(\cdot))$ represents the external force of the B-snake that deforms the contour and moves the snake to the edges of the object. This force is expressed as follows:

$$F(r^0) = -\nabla E_{ext}(r^0) = \left(-\frac{\partial E_{ext}}{\partial x}(r^0), -\frac{\partial E_{ext}}{\partial y}(r^0) \right). \quad (10)$$

The new control points are then obtained as follows:

$$P_i^1 = P_i^0 + \Delta_i^0, \quad \forall i = 0, \dots, M. \quad (11)$$

As a result, the first deformation of the snake curve is described by the following expression,

$$r^1(s) = \sum_{i=0}^M P_i^1 B_{i,3}(s). \quad (12)$$

Then, in the same way we repeat this process. Supposing that we have the k th curve r^k after the k th iteration. Let

$$\Delta_i^k = \mu \left(\int_0^1 F(r^k(s)) B_{i,3}(s) ds + \int_0^1 \sum_{j=0}^M P_j^k \delta_{i,j}(s) ds \right), \quad \forall i = 0, \dots, M \quad (13)$$

$$P_i^{k+1} = P_i^k + \Delta_i^k \quad (14)$$

Thus, the $(k + 1)$ st curve is given as,

$$r^{k+1}(s) = \sum_{i=0}^M P_i^{k+1} B_{i,3}(s). \quad (15)$$

We then generate a sequence of curves $\{r^0, r^1, \dots, r^k, \dots\}$ that describe the deformation of the PID-Snake.

Remark 1. To facilitate implementation, it is advantageous to discretize the adjusting vector given in (13). More precisely, let $\{t_l\}_{l=0}^M$ be a sequence of knots in the interval J . By applying the trapezoidal rule for numerical integration, the discretized form of the adjusting vector is obtained as follows:

$$A_i^k \simeq \mu \left(\sum_{l=0}^M \omega_l \left[F(r^k(t_l)) B_{i,3}(t_l) + \sum_{j=0}^M P_j^k \delta_{i,j}(t_l) \right] \right), \quad \forall i = 0, \dots, M, \quad (16)$$

where $\Omega = (\omega_l)_{l=0}^M$ represent the trapezoidal weights such that $\Omega = (\underbrace{\frac{1}{2}, 1, \dots, 1, \frac{1}{2}}_{M+1 \text{ times}})$.

Lemma 1. The solution to the total energy minimization problem $\nabla E_{B\text{-snake}} = 0$ is obtained by solving the following system:

$$AP + BF = 0, \quad (17)$$

where $F = ((F_x(t_l), F_y(t_l)))_{l=0}^M$, B is the collocation matrix defined in (2), $P = (P_j)_{j=0}^M$ represents the sequence of the control points, and $A = (a_{i,j})_{i,j=0}^M$, with $a_{i,j} = \sum_{l=0}^M \omega_l (\beta_1 B'_{i,3}(t_l) B'_{j,3}(t_l) + \beta_2 B''_{i,3}(t_l) B''_{j,3}(t_l))$.

Proof. Given that the snake is represented as a B-spline curve, the task of finding an optimal curve that minimizes the total energy is converted to determining the optimal control points $P = \{P_j = (X_j, Y_j), j = 0, \dots, M\}$. This is equivalent to solve the following problem:

$$\begin{pmatrix} \frac{\partial E_{B\text{-snake}}}{\partial X_j} \\ \frac{\partial E_{B\text{-snake}}}{\partial Y_j} \end{pmatrix} = \begin{pmatrix} \frac{\partial E_{ext}}{\partial X_j} + \frac{\partial E_{int}}{\partial X_j} \\ \frac{\partial E_{ext}}{\partial Y_j} + \frac{\partial E_{int}}{\partial Y_j} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \forall j = 0, \dots, M. \quad (18)$$

For all $j = 0, \dots, M$, the partial derivatives of the external energy are computed as follows:

$$\begin{aligned} \frac{\partial E_{ext}(r)}{\partial X_j} &= \int_0^1 - \left(\gamma \frac{\partial(\|\nabla I(r(s))\|^2)}{\partial x(s)} ds + \sum_{p=1}^K \alpha_p \int_0^1 \frac{\partial(\|\nabla \Gamma_p(r(s))\|^2)}{\partial x(s)} ds \right) \frac{\partial x(s)}{\partial X_j} ds \\ &= \int_0^1 F_x(s) B_{j,3}(s) ds, \end{aligned} \quad (19)$$

where $F_x(s) = \frac{\partial E_{ext}(r(s))}{\partial x(s)} = -\gamma \frac{\partial(\|\nabla I(r(s))\|^2)}{\partial x(s)} ds - \sum_{p=1}^K \alpha_p \int_0^1 \frac{\partial(\|\nabla \Gamma_p(r(s))\|^2)}{\partial x(s)} ds$, $s \in [0, 1]$.

Similarly, we get

$$\frac{\partial E_{ext}(r)}{\partial Y_j} = \int_0^1 F_y(s) B_{j,3}(s) ds, \quad (20)$$

where $F_y(s) = \frac{\partial E_{ext}(r(s))}{\partial y(s)} = -\gamma \frac{\partial(\|\nabla I(r(s))\|^2)}{\partial y(s)} ds - \sum_{p=1}^K \alpha_p \int_0^1 \frac{\partial(\|\nabla \Gamma_p(r(s))\|^2)}{\partial y(s)} ds$, $s \in [0, 1]$.

On the other hand, by computing the partial derivative of the internal energy in (4) we obtain

$$\frac{\partial E_{int}(r)}{\partial X_j} = \sum_{i=0}^M X_i \int_0^1 \beta_1 B'_{i,3}(s) B'_{j,3}(s) + \beta_2 B''_{i,3}(s) B''_{j,3}(s) ds, \quad \forall j = 0 \dots, M. \quad (21)$$

In the same way, we obtain:

$$\frac{\partial E_{int}(r)}{\partial Y_j} = \sum_{i=0}^M Y_i \int_0^1 \beta_1 B'_{i,3}(s) B'_{j,3}(s) + \beta_2 B''_{i,3}(s) B''_{j,3}(s) ds, \quad \forall j = 0 \dots, M. \quad (22)$$

By discretizing the expressions (19), (20), (21) and (22) using the trapezoidal numerical quadrature formula and substituting them into Eq. (18), we obtain the following system:

$$\begin{cases} \sum_{l=0}^M \omega_l F_x(t_l) B_{j,3}(t_l) + \sum_{i=0}^M X_i \sum_{l=0}^M \omega_l (\beta_1 B'_{i,3}(t_l) B'_{j,3}(t_l) + \beta_2 B''_{i,3}(t_l) B''_{j,3}(t_l)) = 0 \\ \sum_{l=0}^M \omega_l F_y(t_l) B_{j,3}(t_l) + \sum_{i=0}^M Y_i \sum_{l=0}^M \omega_l (\beta_1 B'_{i,3}(t_l) B'_{j,3}(t_l) + \beta_2 B''_{i,3}(t_l) B''_{j,3}(t_l)) = 0, \end{cases} \quad \forall j = 0, \dots, M. \quad (23)$$

The system (23) can be expressed in matrix form as follows:

$$\mathbf{A}\mathbf{P} + \mathbf{B}\mathbf{F} = \mathbf{0}. \quad (24)$$

Lemma 2. The matrix $\mathbf{A} = (a_{i,j})_{i,j=0}^M$ is non-singular and positive definite.

Before proceeding with the proof of Lemma 2, let us recall the theorem established in [21].

Theorem 3. A matrix $\mathbf{D}$ of rank n is positive definite if and only if there exists a non-singular matrix $\mathbf{C}$ such that $\mathbf{D} = \mathbf{C}^T \mathbf{C}$, where $\mathbf{C}^T$ denotes the transpose of the matrix $\mathbf{C}$.

Proof of Lemma 2. The matrix $\mathbf{A} = (a_{i,j})_{i,j=0}^M$ can be written as $\mathbf{A} = \mathbf{A}_1 + \mathbf{A}_2$, where $\mathbf{A}_1 = \left(\sum_{l=0}^M \omega_l \beta_1 B'_{i,3}(t_l) B'_{j,3}(t_l) \right)_{i,j=0}^M$, and $\mathbf{A}_2 = \left(\sum_{l=0}^M \omega_l \beta_2 B''_{i,3}(t_l) B''_{j,3}(t_l) \right)_{i,j=0}^M$.

Let $\mathbf{C}_1 = \left(\sqrt{\beta_1 \omega_l} B'_{i,3}(t_l) \right)_{l,i=0}^M$ and $\mathbf{C}_2 = \left(\sqrt{\beta_2 \omega_l} B''_{i,3}(t_l) \right)_{l,i=0}^M$ be the weighted collocation matrices of the first and second derivatives of the spline basis, respectively. We have

$$\mathbf{A}_1 = \mathbf{C}_1^T \mathbf{C}_1 \quad \text{and} \quad \mathbf{A}_2 = \mathbf{C}_2^T \mathbf{C}_2.$$

The matrices $\mathbf{C}_1$ and $\mathbf{C}_2$ are non-singular. Indeed, for $(a_0, \dots, a_M) \in \mathbb{R}^{M+1}$, we assume that we have:

$$a_0 \left(\sqrt{\beta_1 \omega_0} B'_{0,3}(t_0), \dots, \sqrt{\beta_1 \omega_M} B'_{0,3}(t_M) \right) + \dots + a_M \left(\sqrt{\beta_1 \omega_0} B'_{M,3}(t_0), \dots, \sqrt{\beta_1 \omega_M} B'_{M,3}(t_M) \right) = \mathbf{0},$$

where $\mathbf{0}$ is the zero vector of size $M+1$.

Then we get,

$$\left(\sqrt{\beta_1 \omega_0} \sum_{i=0}^M a_i B'_{i,3}(t_0), \dots, \sqrt{\beta_1 \omega_M} \sum_{i=0}^M a_i B'_{i,3}(t_M) \right) = \mathbf{0}.$$

For instance, consider the first equation. By using the derivative property of spline basis functions, we obtain:

$$\frac{a_0}{h_k} B_{0,2}(t_0) + \sum_{i=1}^M \frac{a_i - a_{i-1}}{h_k} B_{i,2}(t_0) + \frac{a_M}{h_k} B_{M+1,2}(t_0) = 0.$$

Since the B-spline functions are linearly independent, we can therefore conclude that,

$$a_0 = a_1 = \dots = a_M = 0$$

Similarly, we can prove that the matrix $\mathbf{C}_2$ is also non-singular.

Then, from Theorem 3 we deduce that both $\mathbf{A}_1$ and $\mathbf{A}_2$ are positive definite matrices. Furthermore, as the sum of two positive definite matrices yields a positive definite matrix, we therefore deduce that $\mathbf{A}$ is also positive definite.

Theorem 4. The PID-Snake model, which evolves from an initial position in the image using the progressive iterative deformation procedure, converges towards the object boundaries if and only if the parameter μ satisfies the following condition:

$$-\frac{2}{\lambda_n} < \mu < 0, \quad (25)$$

where λ_n is the maximum eigenvalue of the matrix $\mathbf{A}$.

Proof. For all $k \in \{0, 1, \dots\}$, let $\mathbf{P}^k = (P_0^k, \dots, P_M^k)^T$ denotes the control point vector after the k th iteration. After discretizing the integral, Eq. (14) can be expressed in matrix form as follows:

$$\mathbf{P}^{k+1} = \mathbf{P}^k + \mu (\mathbf{A}\mathbf{P}^k + \mathbf{B}\mathbf{F}). \quad (26)$$

To demonstrate the convergence of the proposed progressive iterative deformation procedure, we prove that it converges to the solution of the total energy minimization problem. To do so, we compute the difference between the solution described in Eq. (24) and $\mathbf{P}^{k+1}$. Denoting the identity matrix of rank $M+1$ as $\mathbf{I}$, we obtain:

$$\begin{aligned} \mathbf{P}^{k+1} + \mathbf{A}^{-1} \mathbf{B} \mathbf{F} &= \mathbf{P}^k + \mu (\mathbf{A}\mathbf{P}^k + \mathbf{B}\mathbf{F}) + \mathbf{A}^{-1} \mathbf{B} \mathbf{F} \\ &= (\mathbf{I} + \mu \mathbf{A}) (\mathbf{P}^k + \mathbf{A}^{-1} \mathbf{B} \mathbf{F}) \\ &= (\mathbf{I} + \mu \mathbf{A})^2 (\mathbf{P}^{k-1} + \mathbf{A}^{-1} \mathbf{B} \mathbf{F}) \\ &\vdots \\ &= (\mathbf{I} + \mu \mathbf{A})^{k+1} (\mathbf{P}^0 + \mathbf{A}^{-1} \mathbf{B} \mathbf{F}). \end{aligned} \quad (27)$$

Let $\{\lambda_i, i = 0, \dots, n\}$ be the eigenvalues of the matrix $\mathbf{A}$, sorted in non-decreasing order. Then, the eigenvalues of the matrix $\tilde{\mathbf{A}} = (\mathbf{I} + \mu \mathbf{A})$ are given by $\tilde{\lambda}_i = 1 + \mu \lambda_i, \quad \forall i = 0, \dots, n$.

Algorithm 2: Progressive iterative deformation procedure of the snake model

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input : Input color or gray image  $I$ .
        Initial control points  $P^0 = \{P_i^0\}_{i=0}^M$ .
output: Final PID-Snake curve  $r^{final}(s)$ , where  $s \in [0, 1]$ .
1 //Snake initialization
2    $r \leftarrow r^0$ 
3 // External force computation
4    $F \leftarrow \nabla(E_{ext})$ 
5    $\delta_{i,j}(s) \leftarrow \beta_1 B'_{j,3}(s) B'_{i,3}(s) + \beta_2 B''_{j,3}(s) B''_{i,3}(s)$ ,  $i, j = 0, \dots, M$ 
6 // Progressive iterative deformation process
7    $k = 0$ ;  $n_{iteration} = 0$ 
8   while  $\nabla E_{ext}(r^k) \neq 0$  do
9      $\Delta_i^k \leftarrow \mu \left( \sum_{l=0}^M \omega_l \left[ F(r^k(t_l)) B_{i,3}(t_l) + \sum_{j=0}^M P_j^k \delta_{i,j}(t_l) \right] \right)$ ,  $i = 0, \dots, M$ 
10     $P_i^{k+1} \leftarrow P_i^k + \Delta_i^k$ ,  $i = 0, \dots, M$ ,
11     $r^{k+1}(s) \leftarrow \sum_{i=0}^M P_i^{k+1} B_{i,3}(s)$ 
12     $k \leftarrow k + 1$ 
13     $n_{iteration} \leftarrow n_{iteration} + 1$ 
14 end
15 return ( $r^{final}$ )

```

Since $\mathbf{A}$ is non-singular and positive definite, we have $0 < \lambda_0 < \lambda_1 < \dots < \lambda_n$. Moreover, supposing that we have $-\frac{2}{\lambda_n} < \mu < 0$, then we get $-2 < \lambda_i \mu < 0$ for all $i = 0, \dots, n$. Therefore we obtain

$$-1 < \tilde{\lambda}_i < 1, \quad i = 0, \dots, n.$$

As a result, the spectral radius of $\tilde{\mathbf{A}}$ noted $\rho(\tilde{\mathbf{A}})$ is bounded such that $0 < \rho(\tilde{\mathbf{A}}) < 1$, consequently we get

$$\lim_{k \rightarrow \infty} \tilde{\mathbf{A}}^k = \mathbf{0}_{\mathcal{M}_{M+1}},$$

where $\mathbf{0}_{\mathcal{M}_{M+1}}$ denotes the zero matrix of size $M + 1$.

Hence, from Eq. (27), we conclude that

$$\begin{aligned} P^\infty + \mathbf{A}^{-1} \mathbf{B} \mathbf{F} &= \tilde{\mathbf{A}}^\infty (P^0 + \mathbf{A}^{-1} \mathbf{B} \mathbf{F}) \\ &= \mathbf{0}_{\mathcal{M}_{M+1,2}}. \end{aligned}$$

Then,

$$P^\infty = -\mathbf{A}^{-1} \mathbf{B} \mathbf{F}.$$

As a result, we get :

$$\mathbf{A} P^\infty + \mathbf{B} \mathbf{F} = 0. \quad (28)$$

Consequently, the PID-Snake model converges to the optimal energy position satisfying (24), which aligns with the object boundaries. This concludes the proof.

Remark 2. In the case where the PID-snake is sufficiently smooth, as is the case with our snake, because the spline curve $r(s)$ is of class C^2 on $[0, 1]$, we eliminate the internal energy functional in the total energy expression. Therefore, the adjusting vector Δ_i^k can be expressed for all $i = 0, \dots, M$ as follows:

$$\Delta_i^k = \mu \int_0^1 F(r^k(s)) B_{i,3}(s) ds. \quad (29)$$

As a result, the PID-snake deforms to minimum energy, driven by the external force.

The general algorithm of the proposed PID process for the deformation of the snake is described in detail in the Algorithm 2.

3. Experiments

In this section, we perform experiments to demonstrate the effectiveness of our proposed PID-Snake model. We examine the influence of the proposed energy functional components (7) on the three classes of images categorized in Table 1. We also illustrate segmentation results and provide computational cost of the progressive iterative deformation process (PID). In addition, we evaluate the quality of the segmentation results using different metrics which are commonly used to check the performance of image

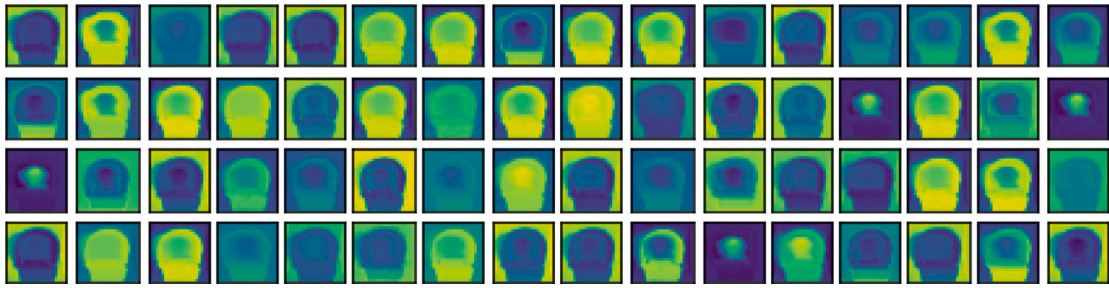


Fig. 1. Representative examples of feature maps (64 channels) extracted from the U-Net model.

segmentation algorithms. Accuracy measures correct predictions divided by total predictions. Sensitivity calculates the ratio of true positives to all positives. Intersection over Union (IoU) and Dice Score evaluate the overlap between segmentation and Ground Truth (GT). The four metrics range from 0 to 1, where a high score indicates a high level of agreement between the segmented region and the GT, while a low score indicates a poor overlap between the two sets.

To implement the segmentation of hard quality images using the U-Net based features [1], we use Nvidia TESLA A100 GPUs with 40 GB of memory and PyTorch v1.8.1 to process the data. Additionally, to train the UNet model, the Adam optimizer was used with a learning rate of 10^{-4} and a batch size of 16. All images were resized to 256×256 . 72% of the samples were used for training, 13% for validation, and the remaining 15% for testing.

Fig. 1 presents representative feature maps produced by the trained U-Net model. These maps are used as inputs to the energy functional that governs the snake deformation process for HC images. For further details, we refer the reader to [7], which provides a comprehensive description of the B-snake energy functional based on the U-Net model.

3.1. Datasets

Using the description outlined in Table 1, we conduct experiments on three dataset types, each representing different levels of image complexity (Low, Medium and Hard), as part of the external energy evaluation process outlined in Algorithm 1. Initially, the creation of a LC dataset is straightforward, involving the collection of real images alongside basic geometric shapes such as circles, cubes, hexagons and so on. On the other hand, the MC dataset was generated using texture images from [22]. We curated a variety of texture classes and paired each texture with another using binary masks of various shapes. Additionally, we incorporated pre-existing composite images into the dataset. This results in a dataset of (256×256) pixel images containing two different classes. Significantly, the segmentation of shapes in these images is based on texture variations rather than changes in intensity color.

Finally, two medical datasets are employed to perform the segmentation of HC images. The Figshare Brain dataset [23], and these Breast Ultrasound dataset [24], each of these datasets consists of images and their corresponding GT labels provided by experts. The Figshare dataset for brain tumor identification contains 3064 MRI images of the brain, including 930 cases of pituitary tumors, 708 cases of meningiomas, and 1426 gliomas. The ultrasound dataset contains breast images from various female patients collected in 2012 at the UDIAT diagnostic center of Parc Tauli Corporation in Sabadell (Spain). This dataset contains 163 images, and each image has abnormalities (53 with cancerous masses and 110 with benign lesions).

On the other hand, for the purpose of comparison with state-of-the-art (SOTA) models (Section 3.5), we use another database known as Drishti-GS [25]. This publicly available dataset consists of retinal images and enables segmentation of the optic disc.

3.2. Ablative study: impact of weights and features energy on the external energy performance

Recall that energy represents intensity changes in the image, with darker regions indicating optimal positions for the snake's energy. These regions influence its deformation and convergence. Therefore, a robust external energy should highlight the object to be segmented with a dark contour, while neglecting other dark regions to prevent the snake from converging towards unintended features and maintaining focus on the desired object. To increase the robustness of the external energy expression (7), the weights γ and $\{\alpha_p\}_{p=1}^K$ play a curious role. This impacts the deformation of the snake and ultimately improves the segmentation results. The use of the best supervised dimensionality reduction technique, which maximizes the Fisher criterion [20], aims to minimize the variance within the object and the background while maximize the variance between them.

To illustrate the impact of the weights γ and $\{\alpha_p\}_{p=1}^K$, as well as the feature energy E_{feat} , on the robustness of the external energy defined in (6), a comparative analysis is presented in Figs. 2 and 3. This comparison highlights the behavior of the energy with and without weighted integration, particularly for images with medium and high segmentation complexity. It is important to note that, in these cases, the feature representations incorporated into the energy functional are derived from Gabor features for the MC dataset and from U-Net feature maps for the HC dataset. When the weight $\gamma = 1$ and the feature energy $E_{\text{feat}} = 0$, the functional energy described in (7) corresponds to the original snake energy [4]. In this situation, as can be seen from Figs. 2 and 3 (second column), the energy performs poorly, particularly in high complexity images where object boundaries are not clearly discernible, while other unwanted details become more pronounced.

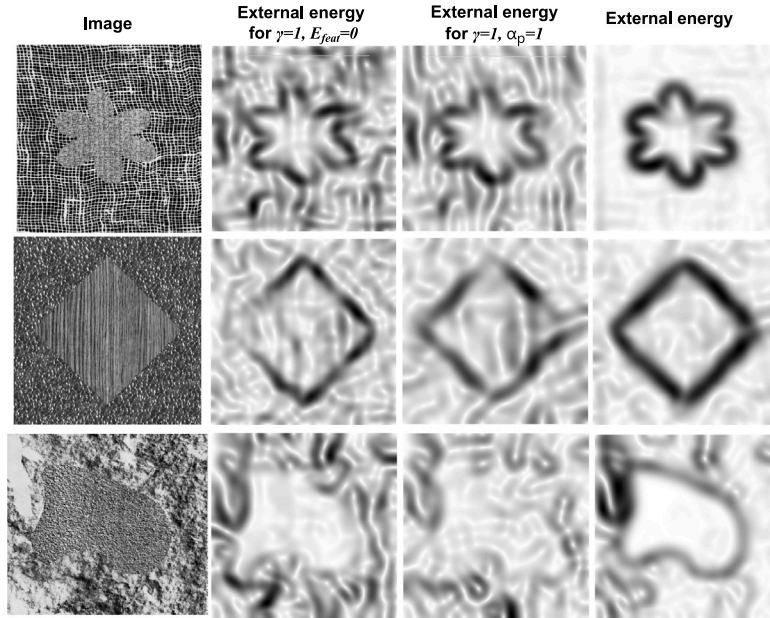


Fig. 2. Illustration of the proposed energy: impact of weights γ , $\{\alpha_p\}_{p=1}^K$, and features energy E_{feat} on Medium complexity images.

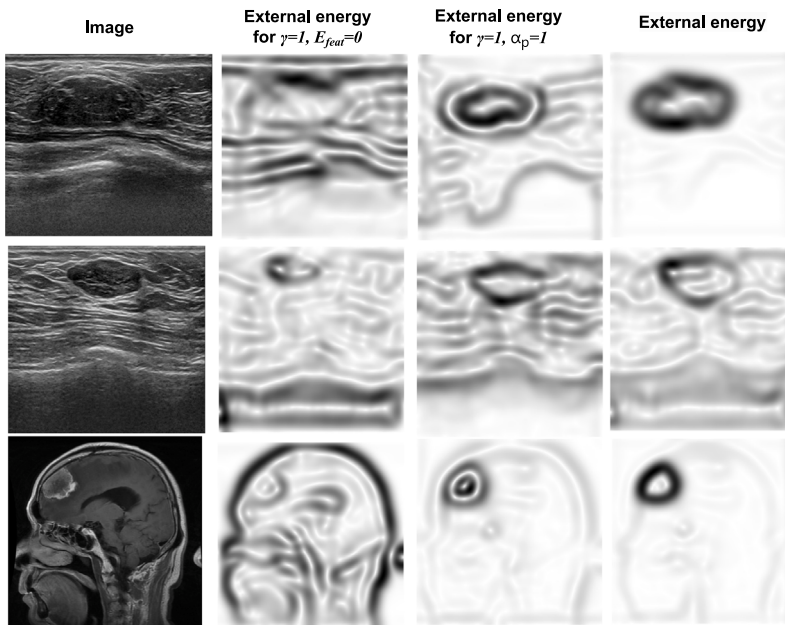


Fig. 3. Illustration of the proposed energy: impact of weights γ , $\{\alpha_p\}_{p=1}^K$, and features energy E_{feat} on Hard complexity images.

Conversely, when $\gamma = 1$ and $\alpha_p = 1$ for $p = 1, \dots, K$ (third column), the feature energy E_{feat} is used without weighting. As observed, these features effectively enhance object boundaries despite challenging image conditions. However, additional irrelevant details may also be emphasized, which can adversely affect the convergence of the snake.

In contrast, the proposed external energy with all its components, as defined in (6) (i.e., the feature energy and the Fisher-based weights), effectively emphasizes the object of interest while suppressing unwanted details. This illustration shows that the feature

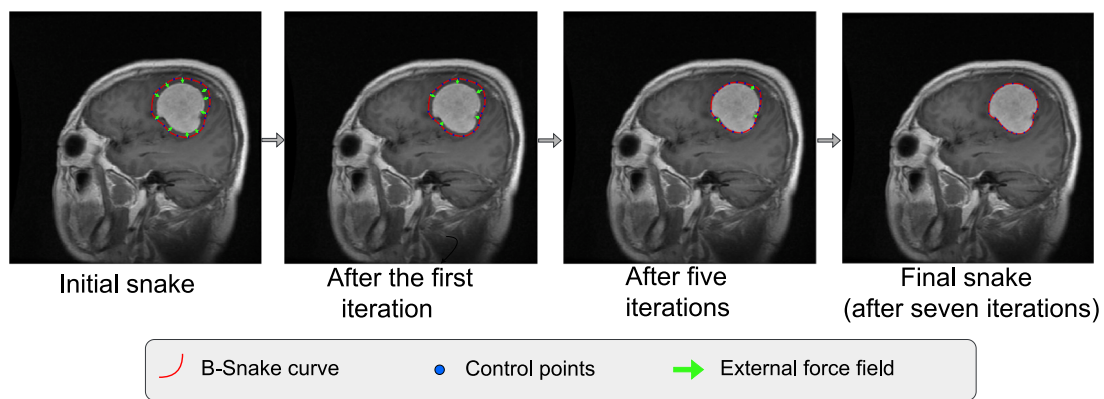


Fig. 4. Illustration of the deformation process of the PID-Snake curve, from its initial position to the object boundaries.

Table 2

Evaluation metrics results using the Figshare Brain dataset.

Figshare Brain dataset	Accuracy	Sensitivity	IoU score	Dice score	Error
Image test 1	0.9981	0.9290	0.9209	0.9608	0.0022
Image test 2	0.9955	0.9950	0.9211	0.9611	0.0052
Image test 3	0.9928	0.9537	0.9172	0.9581	0.0076
Image test 4	0.9972	0.8573	0.8489	0.9251	0.0030

Table 3

Evaluation metrics results using the Breast Ultrasound dataset.

Breast Ultrasound dataset	Accuracy	Sensitivity	IoU score	Dice score	Error
Image test 1	0.9958	0.9538	0.9429	0.9718	0.0045
Image test 2	0.9990	0.9325	0.8998	0.9488	0.0011
Image test 3	0.9870	0.9486	0.9221	0.9608	0.0127
Image test 4	0.9965	0.9632	0.9117	0.9568	0.0039

energy enhances the appearance of the object, whereas the weights improve object–background separation and reduce background contrast.

3.3. Segmentation results

The deformation process of the snake evolves from an initial position until an equilibrium state (i.e., null external force) is reached. Fig. 4 illustrates this iterative evolution of the PID-Snake, starting from the initial position, where the external force progressively attracts the contour towards the object boundary, resulting in convergence after seven iterations.

The segmentation results obtained by using the proposed PID-Snake are shown in Fig. 5 for the three classes of datasets LC, MC and HC. The results confirm the effectiveness of our method as the obtained contours closely match the object contour. This illustration confirms the accuracy and robustness of our model in producing reliable segmentation results for different types of images facing their challenges.

To effectively demonstrate the effectiveness of our model through quantitative results, Tables 2 and 3 present various evaluation metrics for the proposed model applied to the HC images dataset from Figshare Brain [23] and the Breast Ultrasound [24] images, respectively, as both datasets include the corresponding ground truth annotations. It should be noted that the test images corresponding to the quantitative results reported in Tables 2 and 3 are those illustrated in Fig. 5, supplemented with additional samples to ensure a more comprehensive and representative evaluation. The obtained results demonstrate the high performance of the proposed segmentation method, with all four evaluation metrics yielding values close to 1 and associated errors on the order of 10^{-3} .

3.4. Quantitative evaluation of the PID process

To highlight the effectiveness of the proposed PID technique for snake deformation, we conducted a quantitative comparison specifically on the LC dataset. This dataset was selected to enable a direct comparison of our PID process with other deformation techniques, focusing only on its evaluation and using classical energy so as not to influence the results. This comparison involved evaluating some of the computational costs of the proposed method, along with two widely recognized iterative methods for snake

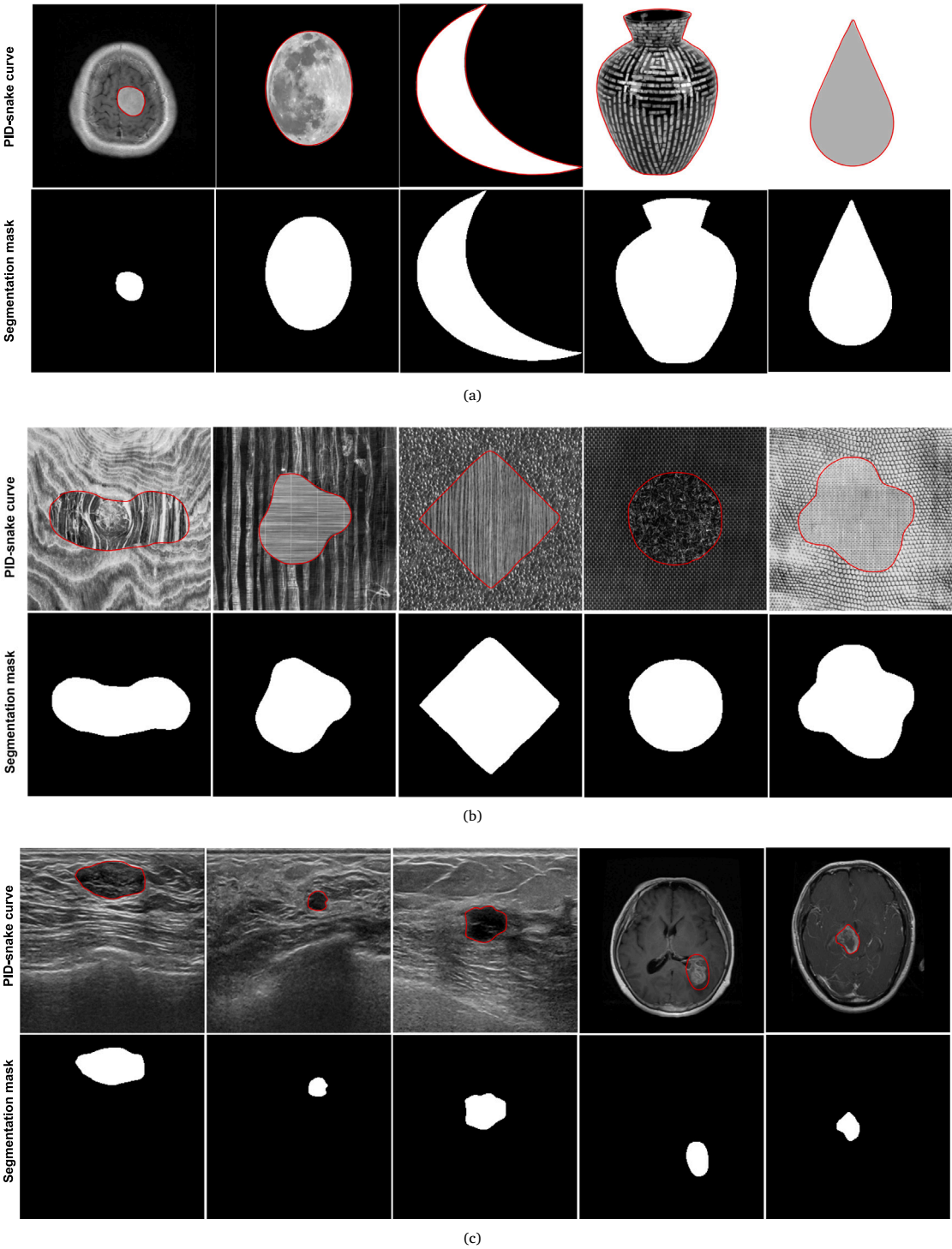


Fig. 5. The segmentation results obtained using the proposed PID-Snake model (shown in red) and their corresponding masks for three dataset classes: LC in (a), MC in (b), and HC in (c).

Table 4

Performance Comparison of the deformation techniques in terms of the time consuming (seconds) and the number of iterations on the LC dataset.

Deformation method	Minimization by SOR		Minimization by SD		Proposed PID process	
	Time (s)	Iterations	Time (s)	Iterations	Time (s)	Iterations
Image test 1	0.3436	20	0.3039	17	0.1357	5
Image test 2	0.3271	19	0.3474	22	0.1295	4
Image test 3	0.3018	23	0.3699	27	0.0992	5
Image test 4	0.2167	15	0.4640	29	0.1492	7

Table 5

Performance comparison of deformation methods: time consumption (seconds) for 7 iterations and Convergence (CV) status on LC dataset.

	Minimization by SOR		Minimization by SD		Proposed PID process	
	Time (s)	CV	Time (s)	CV	Time (s)	CV
Image test 1	0.1895	False	0.1528	False	0.1445	True
Image test 2	0.1558	False	0.1533	False	0.1305	True
Image test 3	0.1379	False	0.1217	False	0.1163	True
Image test 4	0.1773	False	0.1632	False	0.1492	True

Table 6

Quantitative comparison to SOTA recent models using the Drishti-GS dataset.

Method	Accuracy	Sensitivity	mIoU score	mean Dice	Error
Kumar et al. [26]	–	–	0.8310	0.9077	–
Ramani et al. [27]	0.9931	0.9582	–	0.8843	–
Thakur et al. [28]	0.9040	0.8889	0.8224	0.9026	–
Gagan et al. [12]	0.9299	0.9675	0.8410	0.9182	0.0040
Proposed PID-Snake	0.9970	0.9698	0.9165	0.9584	0.0034

deformation: Successive Over Relaxation (SOR) and Steepest Descent (SD). These methods are known for their high convergence speed in solving linear systems of energy minimization.

Table 4, shows the time consuming and the number of iterations to reach the PID-Snake's equilibrium position on the boundaries of the object to be segmented compared to SOR and SD iterative methods. On the other hand, we fix the number of iterations at 7 in order to compare the time required per iteration for each deformation method. Additionally, we provide the status of convergence for the snake, denoted as “True” when convergence is achieved and “False” otherwise. The results are illustrated in Table 5

Observations from Tables 4 and 5 highlight the superiority of the proposed PID process for snake deformation compared to other linear system based techniques, particularly in terms of speed and iteration efficiency. In particular, the PID process demonstrates a notably lower number of iterations required for convergence compared to alternative methods.

The sensitivity of the SOR and SD methods to their parameters has a significant impact on their convergence behavior and the number of iterations required to achieve convergence. In our implementation, we chose to fix the relaxation parameter of the SOR method at 0.5 and the step parameter of the SD method at 0.3 to maintain consistency across experiments. On the other hand, the SOR and SD methods are more computationally costly than the PID method. This difference in computational complexity has a direct impact on the speed of convergence for each iteration.

3.5. Comparison to SOTA models

In this section, we conduct a quantitative comparison of our proposed segmentation model with the latest achievements in state-of-the-art (SOTA) models using the Drishti-GS database [25]. This dataset is widely used in the literature to evaluate the performance of various models, making it a suitable benchmark for comparison purposes. The Drishti-GS database consists of 101 images taken from a population of Indian ethnicity. Each image has a resolution of 2896×1944 pixels. The database provides retinal Optic Disc (OP) images along with corresponding Ground Truth (GT) annotations collected from four experts with varying clinical experience.

The high quality of the images in the Drishti-GS dataset, along with the relatively simple shape of the optic disc, led us to classify it as a low complexity (LC) level. Consequently, we implemented our PID-Snake model without incorporating the energy of features. Fig. 6 presents representative segmentation results obtained with the proposed PID-Snake model on the Drishti-GS dataset. The detected contours, outlined in red, demonstrate a precise alignment with the ground truth boundaries, consistent with the high metric values reported in Table 6.

As shown in Table 6, our PID-Snake model consistently outperforms selected state-of-the-art (SOTA) models cited in [12,26–28], including recent advances in the literature using the Drishti-GS dataset. All comparison metrics for the compared methods are taken from Gagan et al. [12].

The achieved results of 99.7% accuracy, 97% sensitivity, 91.7% mean IoU, and 96% Dice score, with an error rate of 3.4×10^{-3} , are highly promising, affirming the effectiveness of our proposed segmentation model.

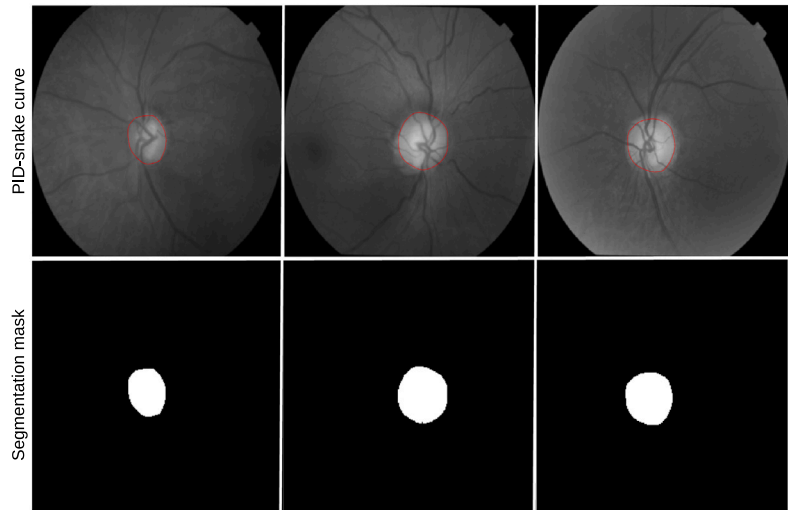


Fig. 6. Representative segmentation results obtained with the proposed PID-Snake model (red contours) and their corresponding masks on the Drishti-GS dataset. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

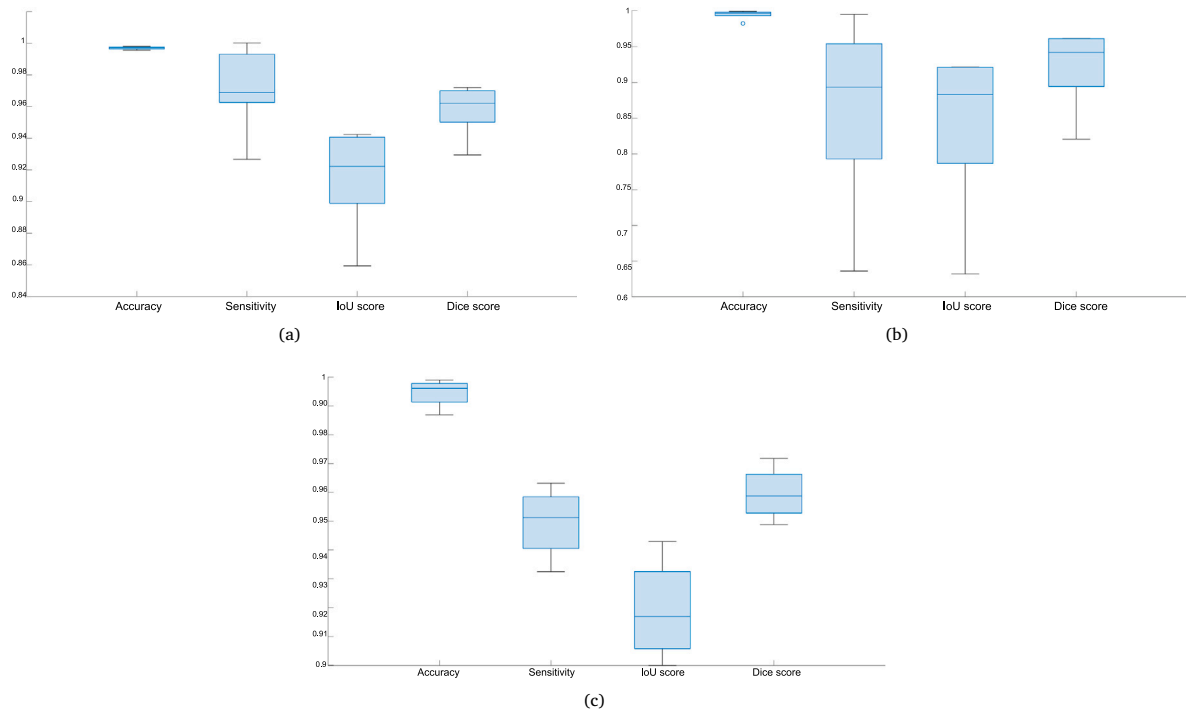


Fig. 7. Distribution of Accuracy, Sensitivity, IoU score, and Dice score for the proposed model throughout Drishti-GS (a), Figshare Brain (b), and Breast Ultrasound databases (c).

Fig. 7 shows boxplots providing a statistical summary of the performance of the segmentation method on the tested datasets, based on accuracy, sensitivity, IoU and Dice score. As can be seen, all of the metrics have high median values, indicating that

the proposed method achieves strong segmentation performance consistently across different images. Additionally, the relatively narrow interquartile ranges reflect low variability in the results, demonstrating the method's stability. The limited spread of the whiskers and the small number of outliers further demonstrate that performance remains reliable even when image quality, noise and structural complexity change. Overall, these observations demonstrate the robustness of the proposed approach, showing that it can maintain accurate and consistent segmentation across different datasets and imaging conditions.

4. Conclusion

In this paper, we have introduced a novel active contour method for image segmentation, known as the PID-Snake method. The proposed model incorporates a robust external energy based on texture and feature analysis, specifically designed to focus on the target object and ensure reliable energy construction across different image types. A key strength of the method lies in its flexibility, as the energy terms can be adapted to the characteristics and complexity levels of the input images, enabling effective segmentation of low, medium and high complexity data.

In addition, we proposed a new deformation strategy, namely the Progressive Iterative Deformation (PID) algorithm, which iteratively adjusts the control points to generate a sequence of fitted curves until the object boundary is accurately delineated. Theoretical and experimental results demonstrate that this approach converges towards the optimal energy configuration without requiring the solution of classical energy minimization problems, which are often computationally expensive and do not guarantee convergence in traditional snake models.

Compared to conventional active contour methods, the proposed PID-Snake model offers improved robustness, reduced computational complexity, and enhanced adaptability to heterogeneous datasets. These properties make it a promising framework for a wide range of applications, particularly in medical image analysis and other vision tasks involving images with varying levels of complexity. Moreover, the proposed approach can be naturally extended to the segmentation of 3D datasets. Overall, this work represents a new direction in snake-based image segmentation, as it departs from classical deformation schemes and traditional energy formulations, opening new perspectives for future research in active contour and deformable models.

CRedit authorship contribution statement

R. Sefti: Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **D. Sbibi:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **R. Jennane:** Validation, Supervision, Resources.

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